

# The Syntax-Prosody Interface: Theoretical Architectures, Metrical Grid Dynamics, and Phrasal Syntax in English

## Theoretical Architecture of the Syntax-Prosody Interface

The interaction between syntactic constituency and phonological phrasing represents one of the most critical structural interfaces in human language. Two major paradigm classes have historically attempted to formalize how syntactic hierarchical trees translate into linear acoustic waveforms: Direct Reference Theories and Indirect Reference Theories<sup>1</sup>. Direct reference frameworks contend that phonological operations can directly inspect syntactic phase structures, constituent boundaries, and c-command relations. Conversely, Indirect Reference Theories—predicated on the Prosodic Hierarchy Theory—posit that phonological rules operate exclusively on an autonomous, hierarchically organized prosodic representation derived from, but not isomorphic to, syntactic trees<sup>1</sup>.

### The Prosodic Hierarchy and Syntax-Prosody Mapping

The classical Prosodic Hierarchy establishes a universal set of bounded prosodic domains arranged in a strict structural containment relationship<sup>1</sup>. From highest to lowest, the phrasal level of the hierarchy comprises:

- **Utterance** ( $\upsilon$ ): The highest prosodic domain, encompassing complete illocutionary acts or multi-clausal turns.
- **Intonational Phrase** ( $\iota$ ): Corresponds to major syntactic clauses (CPs), parentheticals, or dislocated topics, serving as the domain for global pitch contours and boundary tones<sup>3</sup>.
- **Phonological Phrase** ( $\phi$ ): Corresponds to syntactic maximal projections (XPs), serving as the domain for nuclear stress assignment and phrasal sandhi rules<sup>3</sup>.
- **Prosodic Word** ( $\omega$ ): Corresponds to syntactic lexical heads ( $X^0$ ), acting as the domain for word-level stress, root morphophonology, and pitch accent placement<sup>3</sup>.
- **Foot** ( $\Sigma$ ) and **Syllable** ( $\sigma$ ): Lower metrical units governing local rhythmic beat distribution and segmental weight.

Under the traditional Strict Layering Hypothesis, a constituent of category  $C^i$  must be immediately dominated by a constituent of category  $C^{i+1}$ , forbidding both recursive

structures (e.g., a  $\phi$  containing another  $\phi$ ) and level-skipping<sup>4</sup>. However, modern syntax-prosody architectures have revised this stricture, recognizing that prosodic recursion is necessary to reflect recursive syntactic structures like embedded complement phrases or stacked adjuncts<sup>3</sup>. The primary competing models within Optimality Theory (OT) for driving this mapping are Align Theory and Match Theory<sup>2</sup>. Align Theory utilizes edge-alignment constraints (  $\textsc{Align-L/R}$  ) that require the left or right edge of a syntactic constituent (such as an XP) to coincide with the left or right edge of a prosodic domain (such as a  $\phi$ -phrase)<sup>2</sup>. Match Theory instead proposes a strict structural isomorphism through correspondence constraints that directly match syntactic nodes to prosodic nodes<sup>2</sup>.

| Constraint                      | Syntactic Input Domain  | Target Prosodic Output Domain   | Interface Mapping Function and Theoretical Operation                                                      |
|---------------------------------|-------------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------|
| $\textsc{Match}(\text{Clause})$ | Full Clause / CP        | Intonational Phrase ( $\iota$ ) | Maps illocutionary and clausal domains directly to major intonational boundaries <sup>3</sup> .           |
| $\textsc{Match}(\text{XP})$     | Maximal Projection (XP) | Phonological Phrase ( $\phi$ )  | Maps phrasal projections ( $vP$ , $VP$ , $DP$ , $PP$ ) to intermediate prosodic units <sup>3</sup> .      |
| $\textsc{Match}(X^0)$           | Lexical Head ( $X^0$ )  | Prosodic Word ( $\omega$ )      | Maps lexical heads ( $N^0$ , $V^0$ , $A^{\iota}$ ) to individual accentable prosodic words <sup>3</sup> . |

Mismatches between syntactic boundaries and prosodic boundaries arise when these high-ranking  $\textsc{Match}$  constraints are dominated by prosodic markedness constraints<sup>2</sup>. Markedness constraints like  $\textsc{StrongStart}$  (requiring the left edge of

a prosodic domain to begin with a high-prominence element) or eurhythmic length constraints force the prosodic structure to restructure, group, or flatten material into well-formed prosodic units, violating pure isomorphism<sup>2</sup>.

## The Prosodic Status of Functional Categories

A central debate in prosodic phonology concerns the mapping behavior of functional categories (e.g., determiners, auxiliary verbs, prepositions, complementizers) compared to lexical categories<sup>6</sup>. Early formulations asserted that mapping constraints strictly applied to lexical categories while functional heads were systematically ignored<sup>6</sup>. Under this classical

perspective, function words do not project their own  $\omega$ -nodes; instead, they attach as prosodic clitics (prosodic adjuncts or affixes) directly to adjacent lexical prosodic words, as seen in the mapping of [D the] [N cat]  $\implies$  (the (cat) $_{\omega}$ ) $_{\omega}$  or (the cat) $_{\omega}$ <sup>6</sup>.

More recent Match Theory developments establish that `\textscMatch Word` indiscriminately demands that all syntactic heads, both lexical and functional, be mapped to prosodic words<sup>6</sup>. Under this unified framework, the phonological reduction and cliticization of functional items are not caused by syntactic exemption from `\textscMatch`, but rather by lexical subcategorization requirements<sup>6</sup>. Function words possess lexical entries specifying prosodic subcategorization frames that require them to be prosodically defective or unfooted<sup>6</sup>. When these lexical prosodic subcategorization constraints outrank structural

`\textscMatch(X0,  $\omega$ )`, the functional item reduces; conversely, when focus or syntactic movement isolates the functional item, subcategorization requirements are neutralized, and `\textscMatch(X0,  $\omega$ )` forces the functional head to surface as a full prosodic word ( $\omega$ )<sup>6</sup>.

## Metrical Grid Theory, Phonological Metrics, and Stress Assignment Rules

Phonological metrics—as developed by Mark Liberman, Alan Prince, Paul Kiparsky, and Bruce Hayes—model prominence not through isolated phonetic features, but as a relational grid of rhythmic beats<sup>2</sup>. Prominence is determined by the vertical stack of grid marks above syllables within a metrical hierarchy, where Level 0 represents individual syllables, Level 1 designates word-level stress, Level 2 marks phrasal accent, and Level 3 locates the absolute nuclear stress of the intonational phrase.

Phrasal stress placement is historically determined by two primary rules: the Compound Stress Rule (CSR), which assigns primary prominence to the left head in a morphological complex, and the Nuclear Stress Rule (NSR), which assigns primary prominence to the rightmost element within a syntactic constituent:

NSR:  $[XP \alpha \beta] \implies \beta$  receives primary phrasal accent (\*)

## Adjectival Stacking Analysis: "Round the great central life"

Consider the phrase "*Round the great central life*", extracted from classical literary prose<sup>8</sup>. Assuming broad focus and a non-toponymic interpretation, a systematic hierarchy dictates why the adjective "*great*" receives greater prosodic accentuation than the adjacent adjective "*central*".

The sequence "*great central life*" represents a stacked adjectival structure exhibiting asymmetric c-command. The syntax of stacked prenominal adjectives in English is right-branching and non-coordinate. The inner adjective "*central*" forms an immediate nominal constituent with the head noun, as in  $[NP \text{ central life}]$ , functioning as a classifying or domain-restricting modifier. The outer adjective "*great*" merges higher, taking the entire lower NP as its scope:

$[DP \text{ the } [NP [AP \text{ great } [NP [AP \text{ central } [N \text{ life}]]]]]]$

When Match Theory maps this syntactic tree into prosody, the nested syntactic phrases map onto nested, recursive Phonological Phrases ( $\phi$ ):

$(\text{the } (\text{great } (\text{central life})_{\phi})_{\phi})_{\phi}$

The prominence pattern across this domain is derived through the interplay of three distinct metrical mechanisms:

1. **Nuclear Stress Assignment:** Under the Nuclear Stress Rule, primary phrasal stress falls on the rightmost element of the minimal  $\phi$ -phrase, making "*life*" the bearer of the absolute nuclear pitch accent ( $H^*$  or  $L+H^*$ ).
2. **Metrical Grid Resolution and the Rhythm Rule:** Placing primary stress on "*life*" produces a potential metrical clash between adjacent stressed syllables. Both "*great*" and "*central*" carry lexical stress on their initial syllables. The Rhythm Rule (Stress Shift) operates over metrical grids to achieve eurhythm by resolving adjacent stress clashes. To prevent two strong grid marks from occurring in immediate succession before the nuclear accent on "*life*", the metrical beat on "*central*" is relative-demoted or retracts its prominence leftward. However, because "*central*" is trochaic (CEN-tral), its primary stress cannot retract further left. Consequently, the grid mark on "*central*" undergoes metrical

beat deletion or relative reduction to serve as a rhythmic buffer between "great" and "life".

3. **Information-Theoretic Entropy and Functional Prominence:** Information theory models words based on their contextual predictability (entropy  $H$ ). A classifying adjective like "central" exhibits low conditional entropy  $H(\text{Adj} \mid \text{Noun})$  when paired with a head like "life", as it forms a standard conceptual domain. In contrast, the scalar/qualifying adjective "great" carries higher informative entropy and broad semantic scope. Saliency requirements in focus assignment mandate that elements with higher discourse information load receive higher accentual weight.

### The Toponymic Compound Exception

If "Great Central" is interpreted as a compound proper noun or toponym (such as "Great Central Railway"), the underlying syntax is no longer stacked modification, but a lexicalized compound constituent:

$$[\text{NP} [\text{Compound Great Central}] [\text{N life}]] \implies ((\text{GREAT central})_{\omega})_1$$

Under the Compound Stress Rule (CSR), primary accent shifts to the left head of the compound, assigning the primary accent to "GREAT" and reducing "central" to a weak syllable within the compound prosodic word ( $\omega$ ). This structural contrast demonstrates how syntax dictates which metrical rule applies to the grid.

## Sandhi Rules, Segmental Dynamics, and Prosodic Domain Boundaries

Phrasal sandhi processes—segmental mutations that apply across word boundaries—serve as critical empirical diagnostics for mapping the edges of prosodic constituents<sup>1</sup>. These rules do not apply across arbitrary adjacent words; rather, they are bounded by specific nodes of the Prosodic Hierarchy<sup>1</sup>.

- **Flapping and Assimilation:** Bounded internally by the Prosodic Word ( $\omega$ ) or Foot ( $\Sigma$ ).
- **Linking and Intrusive /r/:** Bounded by the Phonological Phrase ( $\phi$ ), applying across adjacent Prosodic Words within that phrase.
- **Sandhi Blocked:** Sandhi rules are strictly blocked across Intonational Phrase ( $\iota$ ) boundaries, where pitch resetting and pause insertion break segmental coarticulation.

### Sandhi Rule Application Dynamics

#### Linking and Intrusive /r/

In non-rhotic varieties of English, word-final non-high vowels (/ə, ɑ:, ɔ:/) trigger /r/-insertion when followed by a vowel-initial word. This rule is strictly bounded by the Phonological Phrase ( $\phi$ ):

$$\emptyset \longrightarrow [r] / V]_{\omega} \quad - \quad [_{\omega} V$$

- **Permitted Sandhi** ( $\phi$ -internal):  $[_{\phi} (\text{far})_{\omega} (\text{away})_{\omega}]_{\phi} \longrightarrow [\text{fɑ:}r\text{ə}'\text{weɪ}]$
- **Blocked Sandhi** ( $\iota$ -boundary):  
 $[_{\iota} \text{Saw Clara.}]_{\iota} [_{\iota} \text{Anna, however...}]_{\iota} \longrightarrow \text{Blocked across } \iota \text{ pause.}$

## Alveolar Flapping

The neutralization of alveolar stops (/t, d/  $\rightarrow$  [r]) occurs when an alveolar stop stands between two vowels, provided the following vowel is unstressed or belongs to an adjacent unaccented word within the same prosodic phrase:

$$/t, d/ \longrightarrow [r] / V \_ V_{[-\text{stress}]}$$

This rule operates freely inside a minimal Phonological Phrase  $\phi$  (e.g., "put it on"  $\rightarrow$  [pʊtɪn]), but is blocked if a syntactic movement or discourse pause inserts an Intonational Phrase

boundary ( $\iota$ ) between the verb and the following constituent.

## Coronal Place Assimilation

Coronal stops (/t, d, n/) assimilate in place of articulation to a following labial or velar onset across prosodic word boundaries within  $\phi$ :

$$/t/ \longrightarrow [p] / \_ ]_{\omega} \quad [_{\omega} C_{[\text{labial}]}$$

For example, the phrase "that boy" /ðæt bɔɪ/  $\rightarrow$  [ðæp bɔɪ] illustrates this assimilation. The strength and likelihood of assimilation scale inversely with the height of the domain

boundary in the prosodic hierarchy: ubiquitous within  $\omega$ , common within  $\phi$ , and rare across  $\iota$ .

## Phase-Based Syntax-Prosody Mapping and Split

# Intransitivity

A foundational discovery in syntax-prosody interactions is the prosodic divergence between the two classes of intransitive verbs: Unergatives and Unaccusatives<sup>9</sup>. Under broad focus conditions, unaccusative structures systematically prefer Nuclear Stress placement on the subject, whereas unergative structures assign Nuclear Stress to the verb<sup>9</sup>:

- **Unergative (Subject base-generated in Spec,vP):** "*A child sneezed.*"  $\longrightarrow$  [A child SNEEZED.]  
[cite: 9]
- **Unaccusative (Subject base-generated as V-complement):** "*A glass fell.*"  $\longrightarrow$  [A GLASS fell.]  
[cite: 9]

Structurally, unergative verbs possess a strong  $vP$  phase boundary containing the subject in  $[\text{Spec}, vP]$ , directing Nuclear Stress onto the verb in the complement of CP<sup>12</sup>. Conversely, unaccusative verbs feature a defective  $vP$  node; the subject originates internally as a V-complement, retaining primary nuclear accent when mapped to PF<sup>9</sup>.

## Phase-Based Prosodic Spellout Mechanics

Modern Phase Theory explains these stress alternations by linking prosodic domain formation directly to the syntactic operation  $\text{Transf}_{\text{PF}}$  at phase boundaries<sup>13</sup>. Syntactic phases ( $CP$  and transitive/unergative  $vP$ ) serve as the spellout domains for phonological evaluation<sup>13</sup>.

When a phase head is merged, its complement is transferred to the Phonological Form (PF) interface, where the Nuclear Stress Rule immediately targets the lowest or rightmost element in that spellout domain<sup>13</sup>.

| Verb Class | Syntactic Phase Architecture                                                                                    | PF Spellout Domain Mechanics                                                                       | Prosodic Output Realization                                                   |
|------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Unergative | $vP$ is a <b>strong phase</b> <sup>13</sup> . Subject is in $[\text{Spec}, vP]$ ; $v$ stays low <sup>12</sup> . | The VP complement is transferred first, then CP targets the raised subject and $v$ <sup>13</sup> . | Nuclear Stress is assigned to the verb within its phase domain <sup>9</sup> . |

|                     |                                                                                                    |                                                                                               |                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Unaccusative</b> | $vP$ is <b>defective</b> (non-phasal) <sup>13</sup> . The first strong phase is CP <sup>13</sup> . | The entire VP and subject undergo a single \textsc{Transfere} operation at CP <sup>13</sup> . | Subject is base-generated as a V-complement, retaining primary nuclear accent <sup>9</sup> . |
|---------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|

### Information Structure and Topic Override

While underlying argument structure sets the default nuclear stress in broad focus, pragmatic Information Structure (IS) can override phase-based accentuation<sup>10</sup>. Discourse factors such as Question Under Discussion (QUD), Contrastive Focus, and Topicality restructure prosodic prominence<sup>10</sup>.

In categorical judgments (where an entity is introduced as a topic about which a predicate comments), the subject projects an independent Intonational Phrase ( $\iota$ ), forcing pitch accents onto both subject and verb regardless of unaccusativity: (The glass) $\iota$  (FELL) $\iota$  <sup>16</sup>. In contrast, inthetic judgments (pure event reporting with no topic-comment structure), the entire clause forms a single  $\iota$ -phrase, allowing underlying syntactic base-generation to dictate stress placement: (A GLÁSS fell) $\iota$  <sup>16</sup>.

### Syntactic Phenomena and Their Mapping to Prosody and Metrics

The following structural mapping matrix synthesizes the operational relationships across core syntactic phenomena, their prosodic realizations, and the governing interface mechanisms.

| Syntactic Phenomenon            | Structural Configuration                                      | Surface Prosodic Realization                                       | Governing Metrical/Interface Mechanism                                             |
|---------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Anaphora / Cataphora</b>     | Coreference chains; given information binding <sup>17</sup> . | Complete deaccentuation of coreferential pronominal or lexical NP. | Givenness / Saliency constraint overrides NSR; prevents default stress assignment. |
| <b>Do-Support / Verum Focus</b> | Stranded Tense affix requiring dummy host under               | High-falling accent ( $H^* +$ ) on auxiliary <i>do</i> or finite   | Focus on Polarity ( $\text{Pol}^0$ ) projects pitch accent onto                    |



|                                     | T <sup>18</sup> .                                                         | V <sup>19</sup> .                                                                               | T-head <sup>19</sup> .                                                                             |
|-------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Differential Object Marking</b>  | Accusative object movement / topicality marking <sup>23</sup> .           | Expanded pitch range and structural separation into isolated $\phi$ -phrase <sup>24</sup> .     | Syntactic shift to Spec,VP forces \textsc{Match-XF} boundary insertion <sup>3</sup> .              |
| <b>Exceptional Case Marking</b>     | Matrix V assigns Case to embedded Spec,TP subject.                        | Embedded subject integrates into matrix verb's Phonological Phrase ( $\phi$ ).                  | Elimination of CP phase boundary allows $\phi$ -domain restructuring <sup>13</sup> .               |
| <b>Right Node Raising (RNR)</b>     | Parallel structures sharing a right-peripheral element <sup>17</sup> .    | Obligatory $\iota$ -boundary pause before shared target; heavy nuclear accent <sup>17</sup> .   | Rightward movement/coordination creates independent Intonational Phrase ( $\iota$ ) <sup>3</sup> . |
| <b>Zero Copula</b>                  | T-node lacks overt phonetic host (e.g., AAVE) <sup>18</sup> .             | Complete suppression of prosodic word node ( $\omega$ ); zero phonetic interval <sup>18</sup> . | Repair-by-deletion under T-strand when c-commanded by overt Agree head <sup>18</sup> .             |
| <b>Inverse Copular</b>              | Predicate Nominal preposed to Spec,TP over Subject.                       | Focal pitch accent shifts to preposed nominal predicate; subject deaccented.                    | Information Structure fronting maps Focus feature to primary pitch accent ( $H^*$ ).               |
| <b>Clausal Stripping / Sluicing</b> | Ellipsis of TP complement leaving WH- or contrast remnant <sup>17</sup> . | Remnant carries isolated nuclear pitch accent; silent phonetic span for                         | PF-deletion of transferred phase domain following focus movement to                                |

|                                       |                                                                    |                                                                                           |                                                                                             |
|---------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
|                                       |                                                                    | TP <sup>27</sup> .                                                                        | Spec,CP <sup>13</sup> .                                                                     |
| <b>Extraposition / Heavy NP Shift</b> | Complex constituent moved to right periphery.                      | Heavy constituent forms an isolated, appended Intonational Phrase ( $\ell$ ).             | Eurhythmic length requirements override default \textsc{Match-XF} alignment <sup>3</sup> .  |
| <b>Subject-Aux Inversion (SAI)</b>    | Auxiliary fronting to $C^0$ head in interrogatives <sup>28</sup> . | Auxiliary undergoes prosodic cliticization unless targeted by Verum focus <sup>19</sup> . | Head movement ( $T^0 \rightarrow$ ) deprives aux of independent phrasal stress.             |
| <b>Topicalization</b>                 | Left-periphery fronting of constituent to Spec,TopP.               | Fronted topic forms independent $\ell$ -phrase with continuation rise ( $L + H^* L -$ ).  | TopP projects explicit CP-level phase boundary mapped to Intonational Phrase <sup>3</sup> . |
| <b>Tough-Movement</b>                 | Null operator movement creating long-distance dependency.          | Primary nuclear stress falls on matrix predicate ("hard"), not shifted object.            | In-situ gap leaves no phonetic target; default NSR applies to matrix VP.                    |

## In-Depth Mechanics of Key Syntactic Phenomena

### Anaphora, Cataphora, and Antecedence

When an NP enters the discourse as a novel referent, it carries high information entropy and receives primary phrasal stress<sup>17</sup>. Once established as an antecedent, subsequent coreferential tokens (anaphora) represent "Given" information. Prosodic interfaces implement a strict Givenness constraint: Given material cannot bear nuclear accent. Consequently, in anaphora, stress shifts leftward to non-anaphoric elements:

Novel: John bought a [BOOK]<sub>Focus</sub>  $\implies$  Anaphoric: Mary [READ]

In cataphora, where the pronoun precedes its antecedent ("*When he<sub>i</sub> entered, John<sub>i</sub> sat down*"), the unindexed pronoun is fully reduced to a clitic, while the postposed antecedent carries full nuclear pitch accent to resolve referential dependency.

### Do-Support, Verum Focus, and Polarity Focus

In English, negation, inversion, or emphatic polarity triggers the insertion of dummy *do* into Tense ( $T^0$ ) when no auxiliary is present<sup>18</sup>. Under standard declarative conditions, tense affixes lower onto the verb<sup>18</sup>. However, when Polarity Focus (Verum Focus) is asserted to contradict a previous denial, a high pitch accent ( $H^*$ ) must target the  $T/Pol$  head<sup>19</sup>:

Polarity Focus Target: [TP John [T DID<sub>H\*</sub>] [VP leave]] vs. Standard: [TP John [VP LEFT<sub>H\*</sub>]]

Because pitch accenting requires a host capable of supporting a Prosodic Word ( $\omega$ ), the stranded Tense node can no longer undergo morphological lowering or deletion<sup>18</sup>. *Do*-support is thus a prosodically driven syntactic repair mechanism that provides a physical host for Verum Focus pitch accentuation<sup>18</sup>. Standard tense lowering combines  $T^0[-past] + VP[leave]$  to form  $(left)_\omega$  bearing nuclear accent on the verb. Under Verum focus, the accentuation of  $T^0[DID]$  with an  $H^*$  accent blocks lowering, yielding  $(DID)_\omega (leave)_\omega$  where primary focus accent falls directly on the auxiliary.

### Differential Object Marking (DOM)

In languages exhibiting DOM (and in English constructions involving shifted or double-object structures), objects marked for high animacy, definiteness, or specificity undergo syntactic movement out of the VP to higher functional positions (e.g., Spec,AgrOP)<sup>23</sup>. Prosodically, unmarked objects stay inside the VP phase and receive default Nuclear Stress under the NSR<sup>14</sup>. DOM objects, having evacuated the VP phase, cross a phase edge<sup>14</sup>. This movement isolates

the object into a distinct Phonological Phrase ( $\phi$ ), triggering higher acoustic intensity, longer duration, and altered pitch contours to signal its structural independence<sup>24</sup>.

### Exceptional Case Marking (ECM) and Raising to Object

In ECM constructions ("*She proved him to be wrong*"), the matrix verb takes a complement TP lacking a CP phase boundary. Syntactically, the embedded subject ("*him*") receives accusative case from the matrix verb and raises to the matrix object position (Raising to Object). Prosodically, the absence of an intervening CP phase allows the raised object to restructure

directly into the matrix verb's Phonological Phrase ( $\phi$ ):

ECM Prosodic Structure: (She proved him) $_{\phi}$  (to be WRONG) $_{\phi}$

This contrasts sharply with full CP complement clauses (*"She proved [that he was wrong]"*), where the CP boundary forces a prosodic phase break ( $\ell$  or  $\phi$ ), preventing the embedded subject from phrasing with the matrix verb.

### Right Node Raising (RNR) and Gapping

Grammatical ellipsis phenomena enforce strict prosodic constraints<sup>17</sup>. In Right Node Raising (*"John cooked, and Mary ate, the BEANS"*), two coordinated clauses share a single right-peripheral object<sup>17</sup>. Syntactically, the shared constituent is either rightward-moved or subject to multidominator coordination. Prosodically, RNR forces an obligatory Intonational

Phrase ( $\ell$ ) boundary immediately before the raised target, accompanied by high pre-boundary lengthening and a pitch pause:

RNR Prosody: (John COOKED) $_{\ell}$  (and Mary ATE) $_{\ell}$  || (the BEANS<sub>Nuclear</sub>) $_{\ell}$

In Gapping (*"John ate beans, and Mary, CHEESE"*), the central verb is deleted under identity<sup>17</sup>. The remaining non-elided remnants (*"Mary"* and *"CHEESE"*) are placed in pitch-contrast with the parallel elements in the first clause<sup>17</sup>. Gapping requires each remnant to receive an independent accent ( $H^*$ ), separated by an internal prosodic boundary where the verb was deleted<sup>17</sup>.

### Zero Copula Constructions

In African American Vernacular English (AAVE) and various creoles, the present-tense copula *be* may surface as zero (*"He my brother"*)<sup>18</sup>. Syntactic analyses establish that zero copula is constrained by prosodic licensing: copula deletion is permitted only in syntactic environments where standard English allows phonological contraction (*"He's my brother"*)<sup>18</sup>.

In licensing environments with a present predicate complement (*"He is happy"*), contraction yields *"He's happy"*, which can further delete to produce the zero copula form *"He happy"*. Conversely, in non-licensing environments involving ellipsis or stranded T tense nodes (*"I know where he is \_\_\_"*), contraction is forbidden (*"I know where he's"*), thereby strictly blocking zero copula deletion (*"I know where he"*).

Under repair-by-deletion accounts, the auxiliary *be* acts as a functional head under T<sup>18</sup>. When T is c-commanded by a full complement phrase, T undergoes phonological reduction and cliticizes onto the subject pronoun<sup>18</sup>. In AAVE, this prosodically unfooted clitic is deleted at PF<sup>18</sup>.

However, in elliptical contexts where the predicate complement is elided (*"I know where he IS \_\_\_"*), the stranded T node must bear nuclear stress to license the ellipsis site via an Agree relation<sup>26</sup>. Because the copula now carries phrasal accent, both contraction and zero copula deletion are strictly blocked<sup>18</sup>.

### Inverse Copular Constructions

In inverse copular structures (*"The cause of the problem IS John"* vs. *"John is the cause of the problem"*), the predicate nominal fronting alters the discourse specification. Standard copular sentences place nuclear stress on the rightmost complement (*"John is the CAUSE"*). Inverse copular sentences reverse this alignment: the preposed predicate nominal acts as an informational topic, causing nuclear focus to shift squarely onto the post-verbal subject noun phrase (*"JOHN"*), signaling identity assertion under inverted syntax.

### Clausal Stripping and Sluicing

Sluicing involves the deletion of an entire TP complement following the fronting of a WH-operator to Spec,CP (*"Somebody left, but I don't know WHO [t<sub>TP</sub>]"*)<sup>17</sup>. Clausal Stripping similarly deletes an entire clause except for a single focal remnant (*"John ate pizza, and MARY too"*)<sup>17</sup>. Syntactically, the focused remnant raises to a peripheral position outside the TP phase<sup>27</sup>. At PF, the remaining TP phase complement undergoes total deletion<sup>27</sup>. Prosodically, this produces extreme asymmetry: the elided span contains zero acoustic duration, while the isolated sluiced remnant (*"WHO"*) absorbs the entire nuclear stress potential of the clause, surfacing with a high-prominence falling contour ( $H^* + L \ L-H\%$ )<sup>27</sup>.

### Shifting (Heavy NP Shift) and Extraposition

Heavy NP Shift and Extraposition postpose complex syntactic constituents to the right edge of the sentence:

Standard: I met [the man who wrote the book]<sub>Heavy</sub> yesterday.

Shifted: I met yesterday [the man who wrote the book]<sub>Heavy</sub>.

This movement is driven by the syntax-prosody interface to optimize eurhythmy and parsing memory<sup>3</sup>. Prosodically, a heavy, embedded constituent located in the middle of a VP creates an asymmetric, nested phonological phrase ( $\phi$ ) that violates the markedness constraint \textsc{EqualSisters} (which favors prosodic phrases of equal weight and duration)<sup>3</sup>. Shifting the heavy clause to the absolute right periphery allows it to map directly into an

independent Intonational Phrase ( $\ell$ ), restoring rhythmic balance and eliminating metrical traps.

### Subject-Verb and Subject-Auxiliary Inversion (SAI)

In Subject-Auxiliary Inversion (SAI), the finite auxiliary moves from  $T^0$  to  $C^0$  ("Can John go?")<sup>28</sup>. Syntactically, the auxiliary vacates the VP phase entirely<sup>28</sup>. Prosodically, because functional heads in  $C^0$  lack independent phrasal projection status unless focused, the fronted auxiliary is deprived of phrasal stress<sup>6</sup>. It cliticizes leftward or rightward as a prosodic prefix to the subject DP:

SAI Prosody: (Can-John) $_{\omega}$  (GO<sub>Nuclear</sub>) $_{\phi}$

When SAI interacts with Subject-Verb inversion in literary fronting ("Down came the RAIN"), the inverted verb moves higher, and the postverbal subject ("the RAIN") lands in final position, receiving default Nuclear Stress as the rightmost element in the phrase under the NSR.

### Movement Paradoxes, Tough-Movement, and Topicalization

Topicalization fronting ("This book, I really LIKE") forces the left-dislocated constituent into Spec,TopP<sup>3</sup>. Prosodically, TopP projects an independent CP-level phase, which Match Theory

maps to an obligatory Intonational Phrase ( $\ell$ )<sup>3</sup>:

Topicalization: (This BOOK) $_{\ell}$  ... (I really LIKE) $_{\ell}$

The topic phrase carries a characteristic rising continuation pitch profile ( $L+H^* L-H\%$ ), signaling that further propositional content follows.

In Tough-Movement ("John is easy to PLEASE"), the surface matrix subject ("John") is syntactically linked to an embedded object gap via a null operator chain. Unlike Topicalization, Tough-Movement does not create a left-peripheral topic phrase. The surface subject integrates directly into the matrix subject DP position, and default Nuclear Stress falls on the embedded verb ("PLEASE"), demonstrating that abstract movement chains without overt dislocated phase heads do not alter default prosodic phrasing.

### Control Structures vs. Unaccusative Movement Traces

Control structures involve an unexpressed null pronoun ( $PRO$ ) in the subject position of an infinitival complement ("John<sub>i</sub> tried [ $PRO_i$  to WIN]"). Because  $PRO$  lacks phonetic features, it projects no node in the metrical grid. Phonological phrasing ignores  $PRO$ , allowing the infinitive marker ("to") to prosodically cliticize directly onto the preceding matrix verb ("tried to")

→ /traɪdtə/).

This contrasts with overt A-movement chains in unaccusatives, where stress reconstruction effects occur<sup>14</sup>. Non-final elements receiving unexpected nuclear stress do so because stress is assigned to the lowermost copy in the VP phase prior to movement<sup>14</sup>. The lower copy sets the phase stress baseline, which reconstructs onto the head of the movement chain at PF<sup>14</sup>.

### Pied-Piping

In Pied-Piping ("*To whom did you speak?*"), a wh-operator drags a governing preposition along with it to Spec,CP. Prosodically, pied-piping prevents the preposition stranding clash where a weak functional preposition is left at the right boundary of a sentence<sup>6</sup>. The fronted

preposition-noun complex forms a consolidated  $\phi$ -phrase at the left edge:

Pied-Piped: (To WHOM) $_{\phi}$  (did you SPEAK) $_{\phi}$

By drawing the functional preposition into the Spec,CP domain alongside the wh-word, Pied-Piping eliminates stranded function words at phrase boundaries, satisfying prosodic alignment constraints at the expense of syntactic minimality.

## Synthesis and System-Wide Conclusions

The overall architecture of the syntax-prosody interface functions as an integrated pipeline:

1. **Syntactic Input Tree:** Syntactic phase derivations ( $CP, vP$ ) construct hierarchical structures containing lexical and functional heads.
2. **Match Theory Mapping Engine:** OT mapping constraints ( $\backslash \text{textscMatch } CP \rightarrow \iota$ ,  $\backslash \text{textscMatch } XP \rightarrow \phi$ ,  $\backslash \text{textscMatch } X^0 \rightarrow \omega$ ) establish structural correspondence between syntactic nodes and prosodic categories<sup>3</sup>.
3. **Prosodic Hierarchy Generator:** The mapped domains form a recursive prosodic hierarchy comprising Intonational Phrases, Phonological Phrases, and Prosodic Words<sup>3</sup>.
4. **Metrical Grid and Eurhythm Filter:** Metrical grid rules (Nuclear Stress Rule, Compound Stress Rule, Stress Shift / Rhythm Rule) resolve stress clashes and enforce rhythmic spacing across the grid<sup>2</sup>.
5. **Acoustic Surface Output:** Segmental sandhi rules (linking /r/, flapping, assimilation) apply strictly within specified prosodic boundaries, generating the final pitch, duration, and intensity variations of natural speech<sup>1</sup>.

The interface between phrasal syntax and prosodic phonology thus operates through a highly structured, constraint-based architecture<sup>2</sup>. Syntactic phase derivations ( $CP, vP$ ) systematically generate the domain windows for phonological evaluation<sup>13</sup>. Match Theory maps these syntactic projections onto the autonomous Prosodic Hierarchy, where surface outputs



are filtered by metrical grid rules, eurhythmmy constraints, and information-theoretic saliency demands<sup>2</sup>. Whether through the stress patterns of adjectival stacks, the phase-driven prominence of unaccusative subjects, or the prosodic licensing of zero copulas and sluiced ellipsis, natural language maintains an absolute alignment between abstract syntactic structure and physical acoustic realization<sup>9</sup>.

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